
Preserving Task-Relevant Geometry in LLM Hidden States: Evidence That Variance-Maximizing PCA Can Obscure Hallucination Trajectory Signals

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Abstract

This study evaluates whether token-level hidden-state trajectories from selected layers of a large language model (LLM) can provide non-destructive signals for detecting response reliability and hedging/hallucination-related behavior [Ji et al., 2023, Azaria and Mitchell, 2023, Farquhar et al., 2024]. The study was conducted under a pre-registered protocol [Nosek et al., 2018, Nakamura, 2026b] in which the primary confirmatory analysis used a 64-dimensional principal component analysis (PCA) projection of the original 3584-dimensional hidden states.

All five Layer 16 representations—PCA 64D (0.7806), PC4–67 (0.7969), BL-LEN (0.8077), raw 3584D PROP- κ (0.8472), and BL-NORM (0.8588)—are compared as *paired* evaluations on the same 98-prompt test inventory ($n = 496$ eligible generations). Under the pre-registered PCA condition, PROP- κ failed the Gate 3 criterion for exceeding the response-length baseline ($\Delta = -0.0271$, $p = 0.8685$; Gate 3: **fail**). Removing compression recovered AUROC = 0.8472 versus BL-LEN ($\Delta = +0.0395$, 95% CI [0.0114, 0.0726], $p = 0.004$; exploratory). Discarding PC1–3 (PC4–67) raised AUROC only to 0.7969 ($\Delta = -0.0108$, $p = 0.6935$). The hidden-state L_2 -norm baseline BL-NORM is the strongest single predictor (0.8588, 95% CI [0.8239, 0.8903]).

PCA diagnostics show that PC1–3 capture 99.933% of calibration variance, consistent with massive-activation dimensions monopolizing global variance in modern LLMs [Sun et al., 2024]. Variance-maximizing PCA therefore collapses the state space to a very low effective rank and can discard directional trajectory geometry. The findings indicate that variance-preserving representation is not equivalent to signal-preserving representation for internal trajectory probing. The load-bearing paired contrast on this split is PC4–67 versus PCA 64D ($\Delta = +0.0163$): discarding the three variance-dominant axes is not a sufficient repair.

Core thesis. The problem is not dimensionality reduction itself; it is reducing the wrong geometry. Variance-preserving dimensionality reduction is not necessarily signal-preserving dimensionality reduction for LLM internal-state trajectory analysis.

Study metadata. Experiment EXP-TRJ001 / EXP-2026-NVS-001. Primary model: Qwen/Qwen2.5-7B-Instruct [Team, 2024, Yang et al., 2024]. Evaluated layers: 8, 16, 24. Hidden dimension $D = 3584$. Primary confirmatory representation: 64D PCA. Paired exploratory and baseline conditions on the same 98-prompt test set: raw 3584D trajectory features, PC4–67, BL-LEN, and BL-NORM.

1 Introduction

1.1 Background and motivation

Large language models have demonstrated strong performance across a broad range of language and reasoning tasks, yet reliable estimation of whether a generated response is correct, supported, or hallucinated remains difficult [Ji et al., 2023, Huang et al., 2025]. Conventional evaluation often observes only the final generated text. Such output-level evaluation is useful, but it provides limited access to the internal dynamics by which a model arrives at an answer.

Hidden states provide a potential non-destructive observation channel into these internal dynamics [Azaria and Mitchell, 2023, Orgad et al., 2024, Marks and Tegmark, 2024, Chen et al., 2024]. During autoregressive generation [Vaswani et al., 2017], the model produces a sequence of internal vector states associated with successive generated tokens. Instead of treating hidden states only as static feature vectors, they can be treated as a trajectory in a high-dimensional representational space [Farquhar et al., 2024].

This study focuses on that trajectory perspective. The central question is not merely whether a hidden-state vector contains information related to response reliability, but whether *the geometry and dynamics of the hidden-state trajectory* contain useful predictive information.

A practical obstacle arises immediately. Hidden states of modern LLMs are high dimensional. The model used in this study has a hidden dimension of 3584. Storing, transporting, and repeatedly processing full trajectories can be computationally expensive. Dimensionality reduction is therefore attractive as an engineering strategy.

However, dimensionality reduction introduces a scientific question that is often treated as an implementation detail:

Which geometry should be preserved?

PCA is a standard answer when the objective is to preserve global variance [Pearson, 1901, Jolliffe, 2002, Jolliffe and Cadima, 2016]. In modern LLMs, however, global variance is often dominated by a few massive-activation dimensions [Sun et al., 2024]. Preserving global variance does not guarantee preserving the geometric structures that encode response reliability.

This study examines that distinction directly.

1.2 The dimensionality-reduction problem

The primary confirmatory protocol specified a 64-dimensional PCA projection. This creates a controlled comparison, executed on one paired 98-prompt test inventory, among (i) uncompressed 3584D hidden-state trajectories, (ii) a variance-maximizing 64D PCA representation and the PC4–67 subspace ablation, (iii) an output-level length baseline (BL-LEN), and (iv) a hidden-state magnitude baseline (BL-NORM).

The underlying concern is that the principal directions that explain the greatest proportion of global variance may preferentially represent broad properties of language generation, while reliability-related dynamics may appear as smaller, localized, anisotropic, or otherwise non-dominant structures.

The point of the present study is not to argue that PCA is intrinsically defective. Rather, it asks whether *variance preservation is sufficient for preserving the particular trajectory signal under study*.

1.3 Research questions

RQ1 — Signal existence and localization. Can token-level hidden-state trajectories from the evaluated layers predict response reliability, and does their discriminative strength vary systematically across the evaluated layers?

RQ2 — PCA signal preservation. Does a variance-maximizing linear PCA projection to 64 dimensions preserve trajectory-based discriminative signal sufficient to outperform the response-length baseline?

RQ3 — High-dimensional recovery. Can trajectory features derived from the uncompressed 3584-dimensional hidden states recover predictive performance beyond the response-length baseline?

1.4 Contributions

This study makes three primary contributions.

1. **Transparent confirmatory negative result.** The pre-registered 64D PCA condition did not satisfy the primary comparative Gate 3 criterion. The failure is reported directly rather than being replaced by a post hoc successful configuration [Simmons et al., 2011, Nosek et al., 2018].
2. **Paired recovery of trajectory signal without variance-max PCA.** On the same 98-prompt test inventory, raw 3584D PROP- κ reached AUROC 0.8472 ($\Delta = +0.0395$ vs. BL-LEN, $p = 0.004$; exploratory). PC4–67 improved on PCA 64D by only $\Delta = +0.0163$ and remained below BL-LEN. BL-NORM (0.8588) is reported in full on the identical $n = 496$ eligible generations.
3. **Massive-activation account of variance collapse.** PC1–3 occupy 99.933% of calibration variance. Following Sun et al. [2024], we interpret this as massive-activation dimensions monopolizing global variance, so that variance-maximizing PCA distorts relative trajectory geometry rather than merely “running out of dimensions.” Intermediate-layer superiority of PROP- κ (Layer 16 vs. 8/24) is reported as a trajectory-curvature reconfirmation of a known mid-depth probing pattern, not as a new localization claim.

2 Background and related work

2.1 Internal-state probing and uncertainty

Probing internal representations for truthfulness and uncertainty has become a standard alternative to output-only evaluation [Belinkov and Glass, 2019, Alain and Bengio, 2017, Zou et al., 2023]. Azaria and Mitchell [2023] showed that linear classifiers on hidden states can detect whether a statement is likely true. Farquhar et al. [2024] introduced semantic entropy to capture meaning-level uncertainty in generated sequences, rather than token-level predictive entropy alone. Related internal detectors such as INSIDE [Chen et al., 2024] and representation-geometry probes [Orgad et al., 2024, Marks and Tegmark, 2024] measure structural properties of intermediate activations. The present work builds on this line by treating the *time series* of hidden states as a trajectory and scoring its discrete curvature κ , rather than a single static probe vector.

2.2 Massive activations and variance dominance

Sun et al. [2024] documented that modern LLMs exhibit *massive activations*: a small subset of hidden dimensions with magnitudes orders of magnitude larger than the remainder, often nearly input-independent and tightly coupled to attention. In any linear variance-based method such as PCA [Pearson, 1901, Hotelling, 1933, Jolliffe, 2002], those dimensions dominate the leading principal components. On Qwen2.5-7B-Instruct, the frozen calibration diagnostic is consistent with that mechanism: PC1–3 capture 99.933% of token-level variance, and PC1–64 capture 99.992%. Variance-maximizing projection can therefore collapse $D = 3584$ into an extremely low effective rank governed by activation *magnitude*, suppressing the subtler directional transitions that trajectory κ is designed to measure.

2.3 Trajectory κ : cosine change and discrete curvature

Let $\mathbf{z}_t \in \mathbb{R}^D$ denote the (possibly PCA-projected) hidden state at generated-token index $t \in \{1, \dots, T\}$ and selected layer ℓ , with $D = 3584$ in raw space and $D = 64$ after the confirmatory projection. Two geometric summaries of successive states are relevant.

Directional (cosine) transition. The one-step cosine change, evaluated after per-token L_2 normalization $\hat{\mathbf{z}}_t = \mathbf{z}_t / \|\mathbf{z}_t\|_2$, is

$$\kappa_t^{\text{cos}} = 1 - \hat{\mathbf{z}}_t^\top \hat{\mathbf{z}}_{t+1} = 1 - \frac{\mathbf{z}_t^\top \mathbf{z}_{t+1}}{\|\mathbf{z}_t\|_2 \|\mathbf{z}_{t+1}\|_2}, \quad t = 1, \dots, T-1. \quad (1)$$

Equation (1) is a *scale-invariant directional* statistic: it is unchanged under any per-token rescaling $\mathbf{z}_t \mapsto \alpha_t \mathbf{z}_t$ with $\alpha_t > 0$. Gram curvature (3) is *not* scale-invariant (it scales as inverse length). Confirmatory PROP- κ remains the frozen Gram estimator; κ^{cos} is the diagnostic that isolates turning from magnitude. Spearman correlation of kappa_mean with mean $\|\mathbf{z}_t\|_2$, κ^{cos} AUROC on L_2 -normalized trajectories, and the residual of κ after regression on norm are the empirical checks of this distinction (scripts/run_trj001_reanalysis.py, Task B).

Frozen confirmatory curvature. The reported PROP- κ scores use the pre-registered discrete curvature of Nakamura [2026b], implemented in src/compute_kappa.py. Parameterization is by token index, not arc length (an explicit, frozen choice: arc-length reparameterization would remove token-to-token speed variation). Central differences are

$$\mathbf{v}_t = \frac{\mathbf{z}_{t+1} - \mathbf{z}_{t-1}}{2}, \quad \mathbf{a}_t = \mathbf{z}_{t+1} - 2\mathbf{z}_t + \mathbf{z}_{t-1}. \quad (2)$$

With Gram matrix G_t of $(\mathbf{v}_t, \mathbf{a}_t)$,

$$\det G_t = \|\mathbf{v}_t\|_2^2 \|\mathbf{a}_t\|_2^2 - (\mathbf{v}_t^\top \mathbf{a}_t)^2, \quad \kappa_t = \frac{\sqrt{\det G_t}}{\|\mathbf{v}_t\|_2^3 + \varepsilon}, \quad (3)$$

where $\varepsilon = 10^{-8}$ is frozen. Endpoints $t = 1$ and $t = T$ are undefined by construction and are excluded. Sequences with $T < 20$ are Class E and are excluded from analysis.

From the valid scalar series $\{\kappa_t\}$, five length-normalized summaries form $\mathbf{x}_\kappa \in \mathbb{R}^5$: mean, maximum, 95th percentile, standard deviation, and AUC density $\frac{1}{T-2} \sum_t \kappa_t$ (kappa_mean, kappa_max, kappa_p95, kappa_std, kappa_auc_density). The confirmatory feature set also includes spike_rate = $(T-2)^{-1} \sum_t \mathbf{1}[\kappa_t > \theta]$, with θ the 90th percentile of train-split κ_t . The raw 3584D ablation dropped spike_rate because train-split raw states were unavailable to calibrate θ ; that condition difference is why ABL-1 is exploratory. A matched 5-feature PCA 64D probe (same summaries, no spike_rate) is the confounder control for feature cardinality and is computed by the reanalysis script (Task D).

2.4 Classifier specification

Every representation—PROP- κ , PROP- κ + LEN, BL-LEN, and BL-NORM—is scored by the same L_2 -regularized logistic regression [Nakamura, 2026b]. Features are standardized with sklearn.preprocessing.StandardScaler fit on the training split only. The inverse-regularization grid is

$$C \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}, \quad (4)$$

selected by validation AUROC and then frozen for the test evaluation. On Layer 16 the freeze record selected $C = 0.1$ for PROP- κ , $C = 0.001$ for BL-LEN, and $C = 100$ for BL-NORM. Test discrimination is AUROC [Hanley and McNeil, 1982, Fawcett, 2006] with prompt-clustered bootstrap (2,000 resamples); DeLong tests are not used because the eight samples of one prompt are not independent [Efron and Tibshirani, 1993].

2.5 PCA versus task-relevant information

For a rank- k projection P_k , the classical PCA objective is conceptually $\max_{P_k} \text{Var}(P_k \mathbf{x})$ under orthogonality [Pearson, 1901, Jolliffe, 2002]. That objective is reconstruction fidelity, not hallucination discrimination:

$$\boxed{\text{high explained variance} \neq \text{high task-relevant information}}. \quad (5)$$

Observation / interpretation / hypothesis. On the paired Layer 16 test split, $\text{AUROC}_{\text{Raw}} = 0.8472 > \text{AUROC}_{\text{PCA64}} = 0.7806$, with $\Delta_{\text{PCA64 vs BL-LEN}} = -0.0271$ ($p = 0.8685$) versus $\Delta_{\text{Raw vs BL-LEN}} = +0.0395$ ($p = 0.004$). The evaluated PCA projection therefore does not preserve all discriminative trajectory information present in the raw states. A candidate mechanism, not established as unique, is that task-relevant directional dynamics are misaligned with massive-activation axes selected by global variance [Sun et al., 2024, Park et al., 2024, Elhage et al., 2022].

Table 1: Frozen experimental environment (results/env_info.json).

Item	Value
Model	Qwen/Qwen2.5-7B-Instruct
Model commit	a09a35458c702b33eeacc393d103063234e8bc28
vLLM	0.29.0
CUDA / driver	CUDA 13.0
Torch	2.13.0+cu130
GPU	NVIDIA A40
Python	3.11.10
Hidden dimension	3584

3 Experimental protocol

3.1 Model and inference environment

The frozen experimental environment records the configuration in Table 1. These values are recorded in the experimental freeze manifest [Nakamura, 2026a]. The model entry explicitly records the migration to Qwen from the previously considered Llama configuration. Inference used vLLM [Kwon et al., 2023].

3.2 Dataset construction and filtering

The experiment uses Type A, Type B, and calibration prompt groups. Phase 1 included 1,000 Type A prompts and 8 samples per prompt, producing 8,000 generations in the pilot stage. The manifest records a mixed-prompt rate of 0.599 and an observed mixed-positive rate of 0.471.

The Phase 2 collection consisted of

$$493 \text{ prompts} \times 8 \text{ samples} = 3,944 \text{ generations.} \quad (6)$$

The recorded label composition is given in Table 2. After mixed-prompt filtering, 299 prompts and 2,392 samples remained. Binary targets map `type_a_positive` to $y = 1$ and hedging/abstention (`type_d_hedge`, `type_c_abstain`) to $y = 0$; the classifiers therefore score reliability versus hedging propensity rather than unhedged factual error alone.

Eligibility is evaluated at the *prompt-cluster* level: a prompt is retained if and only if its eight generations meet the mixed-response criterion, so no partial clusters enter Phase 3 [Nakamura, 2026b]. Prompts were then partitioned at the prompt level into train 296, validation 99, and test 98 (`configs/data_splits.json`).

Paired evaluation set. All Layer 16 comparisons among raw 3584D, PCA 64D, PC4-67, BL-LEN, and BL-NORM are executed on this *same* 98-prompt test inventory. The freeze record reports $n_{\text{test}} = 496$ eligible generations after Class E ($T < 20$) exclusion ($98 \times 8 = 784$ generations; 288 excluded). BL-LEN and BL-NORM are not scored on a different or larger set: they are strictly paired with the trajectory methods on these 496 rows. This pairing removes inventory composition as a confounder: differences are attributable to representation, not to a changing prompt set. Label composition of the 288 Class E rows, and AUROC rank stability under $T \geq 10, 15, 20, 30$, are defined by the reanalysis script and require the freeze trajectory store.

Type A factual prompts are collision-checked with the Wikipedia MediaWiki `action=query` API using the missing flag; no second web-search backend is used (Appendix B).

3.3 Hidden-state extraction

Hidden states are collected at the evaluated layers: Layer 8, Layer 16, and Layer 24. Each token-level state has dimensionality 3584 before projection. The experiment retains both the lower-dimensional PCA trajectory representation and, for the designated ablation subset, the original high-dimensional hidden states.

Table 2: Phase 2 label composition.

Label	Count
type_a_positive	2,191
type_d_hedge	1,616
type_c_abstain	137
Total	3,944

Experimental pipeline: confirmatory PCA vs. exploratory raw trajectories

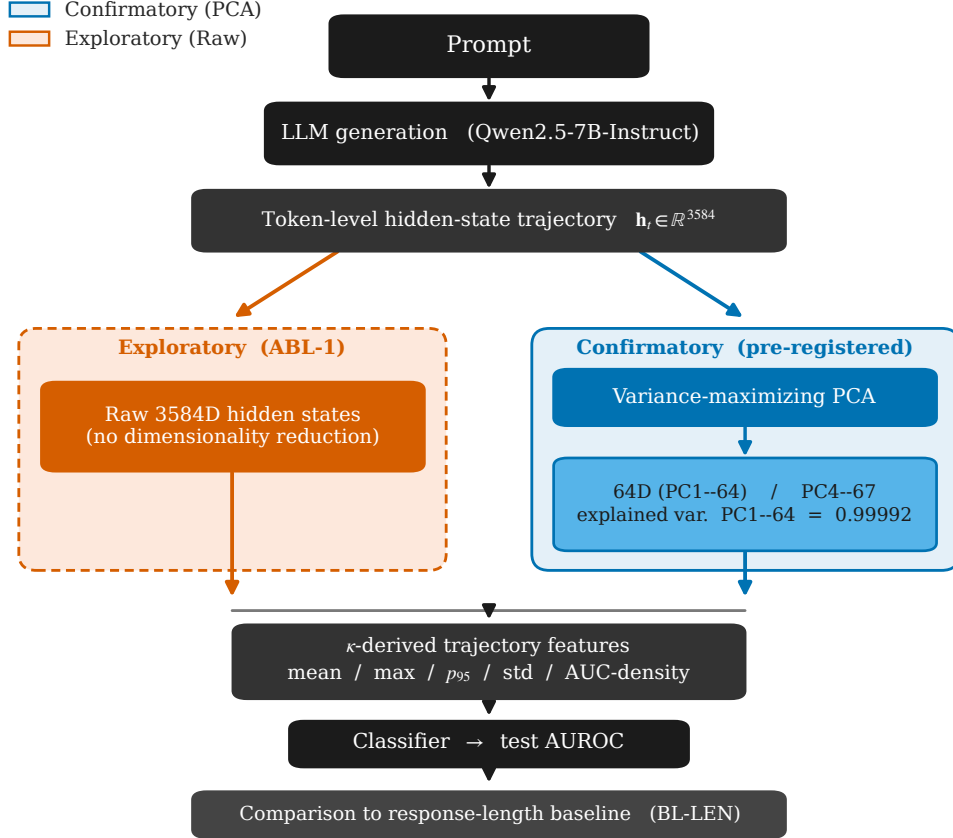


Figure 1: Experimental pipeline. Confirmatory analysis projects token-level hidden states with variance-maximizing PCA before κ -feature construction. The exploratory ABL-1 branch derives the same curvature statistics from uncompressed 3584D states. Both branches, together with BL-LEN and BL-NORM, are scored by the same L_2 -regularized logistic regression on the paired 98-prompt test set.

The manifest records 98 of 200 originally targeted ablation prompts as available and retains that shortfall in `configs/ablation_prompt_ids.json` rather than silently replacing it. Those 98 prompts *are* the paired test inventory used for every Layer 16 representation below.

3.4 PCA configuration

The frozen artifact set contains a PCA projection matrix with shape

$$3584 \times 128. \quad (7)$$

Table 3: PCA diagnostics (Phase 0, Qwen2.5-7B-Instruct).

Diagnostic	Value
PC1–64 cumulative explained variance	0.99992
PC1–128 cumulative explained variance	0.99996
PC1–3 cumulative explained variance	0.99933
Number of calibration tokens	8,357

The PCA diagnostics are reported in Table 3. These values show that the evaluated PCA representation retains extremely high global explained variance while still failing to demonstrate superiority over the response-length baseline in the primary comparative test.

The present study therefore treats explained variance as a representation diagnostic rather than as evidence of downstream task-signal preservation.

3.5 Representations and baselines

Five Layer 16 representations are scored on the paired test set (Figure 1):

1. **PCA 64D (confirmatory)**. Centered hidden states are projected onto PC1–64 (fit on 8,357 calibration tokens) before κ extraction.
2. **PC4–67 (subspace ablation)**. The same protocol on principal components 4 through 67, discarding PC1–3.
3. **Raw 3584D (exploratory)**. κ features from uncompressed hidden vectors; `spike_rate` omitted (no train-split raw θ).
4. **BL-LEN (output baseline)**. Response token length T only. This is the pre-registered Gate 3 comparator.
5. **BL-NORM (magnitude baseline)**. Statistics of $\|\mathbf{z}_t\|_2$ along the trajectory (mean, maximum, standard deviation). Reported in full: it is a first-class baseline, not a suppressed comparator.

PCA diagnostics (Table 3) show that the confirmatory 64D representation retains 99.992% of global variance while still failing Gate 3. Explained variance is therefore a representation diagnostic, not evidence of task-signal preservation.

3.6 Statistical evaluation

The primary discrimination metric is test AUROC. For comparisons against BL-LEN, the analysis reports (i) the AUROC difference Δ , (ii) a bootstrap 95% confidence interval [Efron and Tibshirani, 1993], and (iii) the corresponding empirical comparison p -value reported by the frozen analysis.

Layer-family comparisons use Holm correction [Holm, 1979] across the specified pairwise comparisons.

The study maintains a distinction between

1. **confirmatory analysis**, defined by the pre-registered protocol; and
2. **exploratory analysis**, performed after the pre-registered outcome was obtained.

This distinction is maintained throughout the Results and Discussion sections.

3.7 Pre-registration and freeze protocol

The freeze manifest defines a strict rule [Nakamura, 2026a]:

Data generated before the freeze is not to be adopted as an experimental result.

The manifest also lists hashes for the experimental plan, PCA matrix, centering vector, prompt inventories, label logic, generation configuration, masking procedure, and Type B reference assets.

Table 4: Layer-wise confirmatory PROP- κ test AUROC with 95% bootstrap confidence intervals.

Layer	PROP- κ AUROC	95% CI
8	0.7090	[0.6554, 0.7599]
16	0.7806	[0.7298, 0.8232]
24	0.7167	[0.6570, 0.7693]

Table 5: Paired Layer 16 comparison on the 98-prompt test inventory ($n = 496$). Δ and p versus BL-LEN are cluster-bootstrap estimates from the freeze record where available. BL-NORM Δ is the difference of point estimates; its AUROC CI lies entirely above the BL-LEN point estimate.

Representation	AUROC	Δ vs. BL-LEN	p	95% CI
PCA 64D (confirmatory)	0.7806	-0.0271	0.8685	[0.7298, 0.8232]
PC4-67 (ablation)	0.7969	-0.0108	0.6935	[0.7498, 0.8390]
BL-LEN (output baseline)	0.8077	—	—	[0.7632, 0.8475]
Raw 3584D PROP- κ (exploratory)	0.8472	+0.0395	0.004	[0.8060, 0.8824]
BL-NORM (magnitude baseline)	0.8588	+0.0511	—	[0.8239, 0.8903]

The current uploaded manifest still shows the freeze timestamp, freeze commit, and approver fields as unfilled. The document also specifies that the freeze becomes formally established when all hash fields are complete and the `v1.4-frozen` tag has been applied. This administrative state must be checked and finalized before submission as a formally frozen study.

4 Results

4.1 Confirmatory Gate 3 evaluation (RQ2)

The primary confirmatory evaluation used PCA 64D PROP- κ at Layer 16 on the paired 98-prompt test set:

$$\text{AUROC}_{\text{PROP-}\kappa} = 0.7806, \quad \text{AUROC}_{\text{BL-LEN}} = 0.8077, \quad (8)$$

with $\Delta = -0.0271$, 95% CI $[-0.0727, 0.0190]$, $p = 0.8685$. Gate 3 condition 1 ($\text{AUROC} > 0.5$) passed; condition 2 (exceed BL-LEN) failed:

$$\boxed{\text{Gate 3 = FAIL}}. \quad (9)$$

This negative result is the principal confirmatory finding. It does *not* imply that hidden-state trajectories contain no useful information; it establishes only that the evaluated PCA trajectory representation did not beat BL-LEN under the pre-registered criterion.

4.2 Layer localization (RQ1)

Confirmatory PROP- κ AUROC varies across the three sampled layers (Table 4, Figure 2): 0.7090 at Layer 8, 0.7806 at Layer 16, and 0.7167 at Layer 24. Holm-corrected pairwise tests [Holm, 1979] give Layer 16 vs. Layer 8 $\Delta = +0.0717$ (95% CI $[0.0359, 0.1080]$, $p < 0.001$) and Layer 16 vs. Layer 24 $\Delta = +0.0639$ (95% CI $[0.0368, 0.0944]$, $p < 0.001$). This is localization *within the evaluated set*, not a claim that Layer 16 is globally optimal. When PROP- κ is concatenated with length, Layer 24 reaches 0.8314 versus Layer 16’s 0.8293, indicating complementary sequence-length information in deeper layers.

4.3 Paired comparison of five representations (RQ3)

Table 5 and Figure 3 report every Layer 16 condition on the *same* 98-prompt test set. No comparator is omitted.

Raw 3584D. Removing PCA yields exploratory AUROC = 0.8472 ($\Delta = +0.0395$ vs. BL-LEN, $p = 0.004$). This is not a confirmatory Gate 3 success.

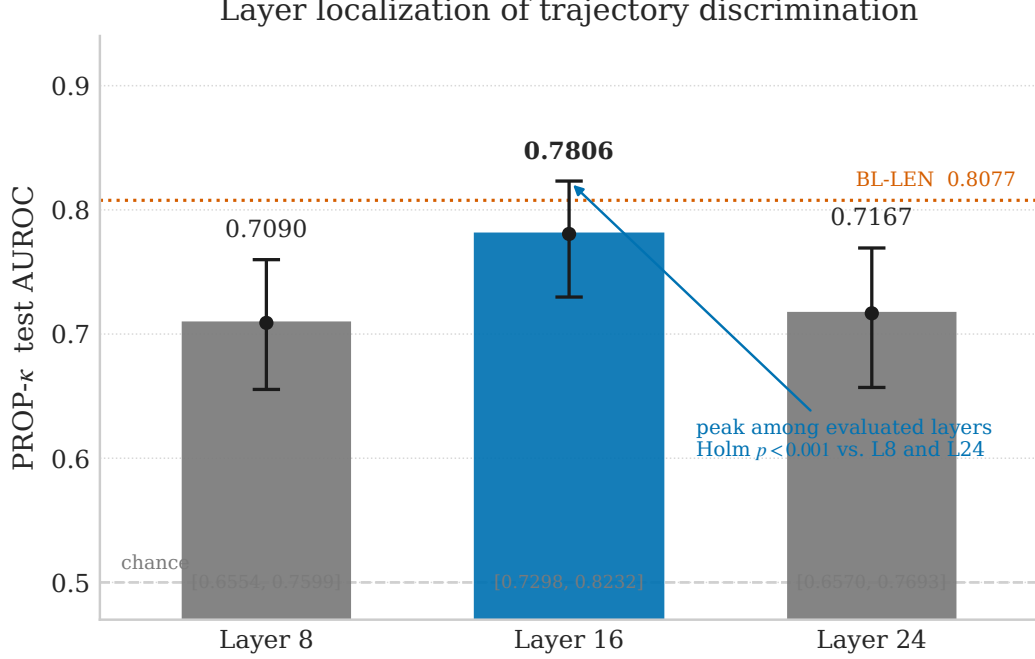


Figure 2: Layer localization of PROP- κ discrimination at Layers 8, 16, and 24, with bootstrap 95% confidence intervals. Layer 16 is the peak among the evaluated layers. The dashed line marks BL-LEN = 0.8077.

PC4–67. Discarding PC1–3 improves on PCA 64D by $\Delta = +0.0163$ but remains below BL-LEN (0.7969, $p = 0.6935$ vs. BL-LEN). Leading-component removal is therefore not a sufficient fix.

BL-NORM. Mean hidden-state L_2 norm is the strongest single predictor (0.8588; CI lower bound 0.8239 $>$ AUROC_{BL-LEN}). Magnitude and directional κ are complementary quantities; a joint model is not claimed here because a frozen combined AUROC is not in the Phase 3 record.

The ordering on this split is

$$\boxed{\text{BL-NORM} \geq \text{raw 3584D} > \text{BL-LEN} > \text{PC4-67} > \text{PCA 64D}}. \quad (10)$$

5 Discussion

5.1 Massive-activation dominance of global variance

The result is not that PCA is intrinsically unsuitable for LLM analysis. It is that two objectives need not share axes (Figure 4): variance/reconstruction versus reliability discrimination.

On Qwen2.5-7B, PC1–3 explain 99.933% of calibration variance and PC1–64 explain 99.992%. Following Sun et al. [2024], we interpret this spectrum as massive-activation dimensions monopolizing global variance. Variance-maximizing PCA then collapses $D = 3584$ into a near-rank-3 magnitude subspace and can discard directional κ dynamics.

This is not a claim that 64 dimensions are numerically insufficient. Discrete curvature κ_t is a property of the osculating plane spanned by $(\mathbf{v}_t, \mathbf{a}_t)$: the Gram determinant in (3) is the squared area of that parallelogram, and κ_t is unchanged if the trajectory is isometrically embedded in any ambient dimension $D' \geq 2$. Extra coordinates that are constant, or that merely rescale the same plane, cannot increase the intrinsic curvature. What PCA can do—and what the paired PC4–67 increment of only $+0.0163$ is consistent with—is *rotate and anisotropically rescale relative token placement* so that the projected curve is a distorted image of the original. The defect is distortion of relative configuration under a variance-max objective, not a missing-dimension budget.

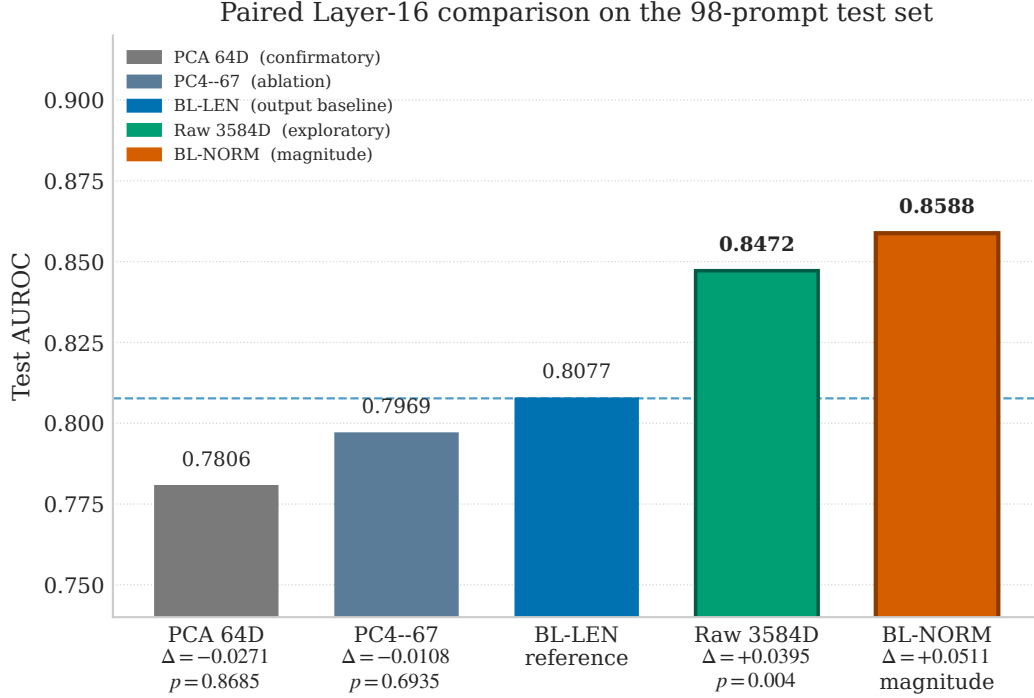


Figure 3: Paired Layer 16 test AUROC on the 98-prompt set: PCA 64D (0.7806), PC4-67 (0.7969), BL-LEN (0.8077), raw 3584D (0.8472), and BL-NORM (0.8588). BL-NORM and PC4-67 are reported in full. Raw vs. BL-LEN is exploratory ($p = 0.004$).

That interpretation remains a mechanism *hypothesis*: the experiment shows signal loss under this projection, not a unique localization of the lost bits in the discarded 0.008% of variance. The same magnitude geometry that dominates PCA is layer-dependent as a predictor: BL-NORM falls from 0.8588 at Layer 16 to 0.7158 at Layer 24, consistent with massive-activation structure changing near the output.

$$\boxed{\text{high explained variance} \not\Rightarrow \text{high task-relevant information}}. \quad (11)$$

5.2 Facing BL-NORM as a first-class baseline

BL-NORM is the strongest individual score on the paired split (0.8588). Hiding it would invite the charge that a stronger internal baseline was suppressed. The result is scientifically informative: hedging/reliability is tightly coupled to activation *magnitude*, exactly the geometry that massive activations and leading PCs occupy [Sun et al., 2024]. Raw κ (0.8472) is a directional statistic; BL-NORM is a magnitude statistic. They are not interchangeable, and Gate 3 remains defined against BL-LEN because that was the pre-registered output-level comparator. A joint κ +norm model is left to follow-up work.

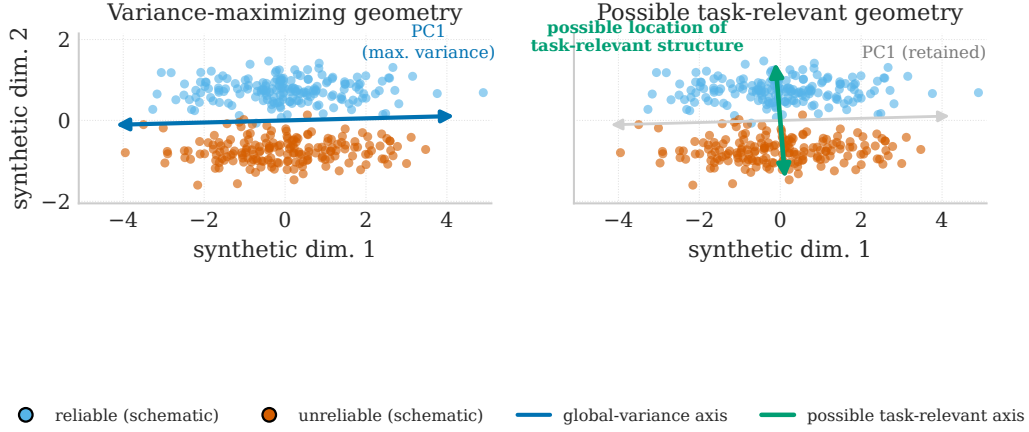
5.3 Facing PC4-67

PC4-67 tests the story that PC1-3 are the only nuisance directions. The paired gain over PCA 64D is small (+0.0163) and PC4-67 still fails to beat BL-LEN (0.7969 vs. 0.8077, $p = 0.6935$). Removing the massive-activation axes is therefore not sufficient to recover the raw-space κ signal (0.8472). Residual explanations—nonlinear structure, local anisotropy, weak distributed directions—remain open [Elhage et al., 2022, Gurnee et al., 2023].

5.4 The central thesis

The study therefore proposes the following design principle:

Global variance is not necessarily the geometry of the task



Schematic only. The experiment does not establish that the signal occupies a specific low-variance subspace.

Figure 4: Schematic: global-variance PCA axes versus *possible* task-relevant structure. The diagram does not claim that the reliability signal occupies a proven low-variance subspace.

The problem is not dimensionality reduction itself; it is reducing the wrong geometry.

A practical consequence is that future projection methods should be evaluated against two different criteria: (i) **representation preservation** and (ii) **task-signal preservation**. A projection can score extremely well on the first and poorly on the second. The present study provides evidence that this distinction matters for LLM hidden-state trajectory analysis.

5.5 Engineering implications

The results matter beyond this particular experiment because high-dimensional hidden states are expensive to retain and process. A future real-time system may need to transform

$$3584D \rightarrow 16D-64D \quad (12)$$

while preserving the trajectory signal relevant to confidence or reliability estimation [Kadavath et al., 2022, Farquhar et al., 2024].

The implication is not that raw 3584D representations should always be retained in production. The more useful engineering question is:

Can a compact learned geometry preserve the same task-relevant structure that the raw representation contains?

This motivates task-aware projection rather than variance-only compression.

5.6 Limitations

5.6.1 Exploratory status of ABL-1

Raw 3584D PROP- κ was obtained after the confirmatory outcome and dropped `spike_rate`; it is exploratory.

5.6.2 Target definition

Binary labels map factual-positive versus hedge/abstain. The classifiers therefore probe hedging propensity, not unhedged factual hallucination alone.

5.6.3 Limited layer sampling

Only Layers 8, 16, and 24 were evaluated.

5.6.4 Limited ablation inventory

98 of 200 designated ablation prompts were available; that shortfall is retained.

5.6.5 Single-model evaluation

Results are for Qwen/Qwen2.5-7B-Instruct only.

5.6.6 Mechanism not identified

Massive-activation dominance of PC1–3 is a supported interpretation, not a unique geometric localization of the lost κ signal.

6 Conclusions and future work

6.1 Conclusion

This study began with a pre-registered hypothesis concerning whether a 64D PCA representation of LLM hidden-state trajectories could exceed a response-length baseline in hallucination/reliability discrimination.

The primary confirmatory analysis did not meet the Gate 3 criterion ($\text{AUROC}_{\text{PCA64}} = 0.7806 < \text{AUROC}_{\text{BL-LEN}} = 0.8077$, $\Delta = -0.0271$, $p = 0.8685$). On the same 98-prompt test set, PC4–67 reached only 0.7969, raw 3584D PROP- κ reached 0.8472 ($p = 0.004$ vs. BL-LEN; exploratory), and BL-NORM reached 0.8588. PC1–3 occupy 99.933% of calibration variance, consistent with massive-activation monopoly of global variance [Sun et al., 2024].

The resulting scientific interpretation is deliberately narrower than a claim that PCA is universally harmful:

The evaluated variance-maximizing PCA projection did not preserve all of the task-relevant trajectory signal available in the raw hidden-state representation.

This supports the broader design principle

$$\boxed{\text{variance-preserving reduction} \neq \text{signal-preserving reduction}} \quad (13)$$

and motivates the search for projection methods that explicitly preserve task-relevant geometry.

6.2 Future directions

6.2.1 Geometry-preserving supervised projection

The next experimental stage should construct a learned projection

$$\mathbb{R}^{3584} \rightarrow \mathbb{R}^{16-64} \quad (14)$$

that optimizes task-relevant structure rather than global variance alone (follow-up experiment TRJ002a). Candidate approaches include supervised linear projection, supervised autoencoders, metric learning, contrastive projection, and low-rank learned projection. The evaluation should compare both predictive performance and geometric distortion.

6.2.2 Dense layer sweep

Instead of testing only Layers 8, 16, and 24, future work should evaluate a substantially denser set of layers to determine whether Layer 16 represents a genuine intermediate-layer maximum or merely a local maximum within the sampled points.

6.2.3 Direct geometry diagnostics

Future experiments should explicitly test candidate mechanisms rather than infer them indirectly. Useful diagnostics include the variance rank of task-discriminative directions, Fisher or between-class / within-class geometry, local neighborhood preservation, curvature and trajectory-shape preservation, and nonlinear separability before and after projection.

6.2.4 Cross-model and cross-dataset validation

The strongest next validation would repeat the experiment across multiple LLM families, multiple parameter scales, independent prompt inventories, and independently curated reliability/hallucination datasets [Kuhn et al., 2023, Manakul et al., 2023].

6.2.5 Real-time integration

A successful compact geometry-preserving projection could eventually serve as a low-latency internal-state signal for real-time control systems such as Project L-Agent and Project V-Agent. The target architecture is conceptually

$$\begin{aligned} \text{LLM hidden state} &\rightarrow \text{geometry-aware projection} \\ &\rightarrow \text{trajectory reliability signal} \rightarrow \text{confidence feedback / control.} \end{aligned} \tag{15}$$

The key requirement is not merely low dimensionality. It is preservation of the information that makes the signal useful.

7 Reproducibility

Artifact hashes, split files, and Phase 1–3 counts are recorded in Appendix A. Prompt inventories and Wikipedia filtering are specified in Appendix B. Detailed AUROC tables, including BL-NORM by layer, appear in Appendix C. Confirmatory statements are those of the pre-registered 64D PCA Gate 3 test; raw 3584D recovery is exploratory; massive-activation dominance of PC1–3 is a mechanistic hypothesis [Nosek et al., 2018, Simmons et al., 2011, Sun et al., 2024]. Administrative freeze fields (timestamp, commit, approver) remain unfilled in the uploaded manifest and are not claimed complete.

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Table 6: Administrative freeze fields. Unfilled tagging fields are recorded as unfilled.

Item	Value
Freeze timestamp (UTC)	<i>unfilled</i> (to be written at tagging)
Freeze commit	<i>unfilled</i>
Tag	v1.4-frozen
Approver	<i>unfilled</i>

Table 7: Environment (measured; results/env_info.json). The model was changed from meta-llama/Meta-Llama-3.1-8B-Instruct under update U8.

Item	Value
Model	Qwen/Qwen2.5-7B-Instruct
Model commit	a09a35458c702b33eeacc393d103063234e8bc28
vLLM	0.29.0
CUDA / Torch	CUDA 13.0 / 2.13.0+cu130
GPU	NVIDIA A40
Python	3.11.10

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A Freeze manifest

Source artifact: FREEZE_MANIFEST.md [Nakamura, 2026a]. The freeze is defined to become formally established when every hash field is populated and the commit is tagged v1.4-frozen. Data generated before that freeze are not to be adopted as experimental results.

The manifest is the authoritative record for environment, artifact hashes, sample counts, data-split definitions, PCA diagnostics, Phase 1–3 outcomes, Gate 3 status, and ABL-1/ABL-2 outcomes.

Table 8: Frozen artifact SHA-256 hashes (measured 2026-09-13). Hashes are wrapped for the column width.

#	Artifact	Path	SHA-256
1	Experimental plan	docs/EXP-2026-NVS-001_v1.4.md	d37d3bc47e5520be628b279c a7d7407ecc0f99a277f9a0eb7bc60b25cf5e0910
2	Projection W (3584×128)	configs/W_pca128.npy	dbfe086fa5dbfc965f28d78d 072f2a6565752c1e19db8a5646403031656e285e
3	Centering vector μ (3584)	configs/mu_pca.npy	521eec467645365eb9e3c9a1 44b751a172ea504ab3acf4f2affed46f39b2d61b
4	Calibration prompts (200)	calibration_prompts_v1.json	492b3a64306bd0f197fa8f75 e0d3391ac260d147ba32e7732e6983e14ba097e7
5	Type A Phase 1 prompts (1,000)	configs/prompts_v1.json	a86d476d395ddb6762f2d862 515778106418b4daec3b149ac6fd5d2e39ba4f06
5b	Type A Phase 2 prompts (493)	configs/prompts_v1_phase2.json	85c6f92d0e9399ad0817099a a2d1429b3f99855813c0fa1d127aa8f8964882cf
6	Refusal / hedging patterns	refusal_patterns_en.json	785419bf0cf5c227ebbae29b 5ee0d997b20f89ffcdcc153ba014a7face488d94
7	Type B reference set	configs/typeB_reference.json	1c4257f0b9b0ed0a97bc3185 f445363dbbf4578d64aea296bb9f1d266e0a86e8
8	κ module	src/compute_kappa.py	bebb8388913062c780d604c3 60ba8b93e7a23afcb3080d8e404870371d860f4b
9	Hallucination label module	src/eval_hallucination.py	5dc9fd88969603ca0271e42d cbf857cc09fcd0c77dbae6b09305263f8b9fbd69
10	Generation config	configs/generation_config.json	30a81eb5746d016c56a8c2fe b47d9b5dbeec7915c6babfb6911b693f56e7d34d
11	Mask protocol	configs/mask_protocol.json	ba08770073ce474cd4cfb436 77751556a2989bb6583e634039d51e06e84bbd2c
12	Type B prompts (200)	configs/prompts_typeB_v1.json	caf7358d36a8022ed1f6db3e fd8481a22994534b7cbec5622be3e1c32709d0fe

Table 9: Phase 1 quantities fixed before Phase 2 generation (2026-09-13).

Item	Value
Measured mixed-prompt rate	0.599 (599/1,000)
Mixed-subset positive rate	0.471
Back-calculated prompt requirement M	444.1 $\rightarrow$ 445 (inventory 493; no extra generation)
Split file configs/data_splits.json	SHA-256 ecd519da...15f17b8f; train 296 / val 99 / test 98
Ablation IDs configs/ablation_prompt_ids.json	SHA-256 78b39383...7736d355; 98/200 available (shortfall field retained)
Phase 1 pilot	1,000 $\times$ 8 = 8,000 generations; Rule 3 pass

Rule 1 note. The 64D (PC1–64) PROP- κ AUROC is not near 0.5; Gate 3 condition 1 passes. The protocol branch that assumed “64D AUROC $\approx$ 0.5 but an ablation is significant” is therefore not strictly triggered. ABL-1 nevertheless exceeds BL-LEN, a pattern the pre-registration did not specifically anticipate: the 64D PCA projection appears to lose κ -derived information that, in raw space, exceeds response length.

B Prompt construction and Wikipedia-based filtering protocol

B.1 Scope and role of this appendix

This appendix specifies the prompt-construction and factual-conflict filtering procedure used in EXP–TRJ001 to the extent supported by the frozen experimental record.

The purpose of this appendix is to make explicit: (i) the relationship among the Type A, Type B, and calibration prompt inventories; (ii) the generation and sampling structure used in the experiment; (iii) the transition from the Phase 1 prompt inventory to the Phase 2 evaluation pool; (iv) the Wikipedia-based factual-conflict filtering stage; and (v) the exact provenance artifacts that determine which prompts and generated samples entered the reported evaluation.

Where a numeric threshold is not a freeze-record field, the appendix cites the operational rule from Nakamura [2026b] rather than leaving a drafting placeholder.

Table 10: Phase 3 confirmatory and ablation outcomes (Layer 16 unless noted).

Item	Value
Mixed prompts after filtering	299/493 (2,392 samples)
PROP- κ test AUROC	0.7806 (95% CI [0.7298, 0.8232])
BL-LEN test AUROC	0.8077 (95% CI [0.7632, 0.8475])
PROP- κ vs. BL-LEN Δ	-0.0271 (95% CI [-0.0727, 0.0190], $p = 0.869$)
Gate 3 condition 1 (AUROC > 0.5, $p < 0.01$)	pass (CI lower 0.730)
Gate 3 condition 2 ($\Delta > 0$, $p < 0.05$)	fail (CI includes 0)
Gate 3 overall	fail
F-Layer (Holm, exploratory)	L16 > L8 and L24 ($p_{\text{holm}} < 0.001$)
ABL-1 (raw 3584D κ , $n = 496$)	AUROC 0.8472; $\Delta = +0.0395$ ($p = 0.004$)
ABL-2 (PC4-67, $n = 496$)	AUROC 0.7969; $\Delta = -0.0108$ ($p = 0.694$)
Metrics path	results/phase3_metrics.json

Table 11: Prompt-related frozen resources.

Resource	Role	Frozen artifact
Calibration prompts	Calibration of the representation / projection procedure	configs/calibration_prompts_v1.json
Type A prompts	Primary inventory for the main data-generation workflow	configs/prompts_v1.json
Type A Phase 2 prompts	Independent inventory for the Phase 2 requirement	configs/prompts_v1_phase2.json
Type B prompts	Reference / verification condition	configs/prompts_typeB_v1.json
Type B reference set	Reference answers / matching rules	configs/typeB_reference.json
Refusal / hedging patterns	Label-related structural patterns	configs/refusal_patterns_en.json

This distinction is important because prompt construction and filtering are part of the experimental condition. A post hoc reconstruction would not be equivalent to the frozen protocol.

B.2 Prompt inventory structure

The experiment uses the prompt-related resources in Table 11. The freeze manifest records SHA-256 hashes for these assets, together with the implementation modules responsible for κ computation and hallucination labeling. These hashes constitute the provenance boundary for the reported study.

B.2.1 Calibration inventory

The calibration inventory contains 200 prompts: configs/calibration_prompts_v1.json. Type A construction uses five equally allocated templates (200 each): fictional_movie, fictional_person, fictional_paper, fictional_chemical, and fictional_location.

B.2.2 Type A inventory

The primary Type A inventory contains 1,000 prompts: configs/prompts_v1.json. The experiment generated 8 samples per prompt during the Phase 1 pilot stage,

$$1,000 \times 8 = 8,000 \text{ generations.} \quad (16)$$

The frozen record reports a mixed-prompt rate of

$$599/1,000 = 0.599 \quad (17)$$

and, within the mixed subset, an observed positive rate of 0.471. These values were used to determine the Phase 2 inventory requirement.

B.2.3 Type A Phase 2 inventory

The Phase 2 inventory contains 493 prompts: configs/prompts_v1_phase2.json. The freeze record states that this inventory was newly constructed as an independent pool and that the calculated

requirement was

$$M = 444.1 \rightarrow 445. \quad (18)$$

A total of 493 prompts was therefore retained, providing a margin above the calculated requirement without additional post hoc prompt generation. The Phase 2 collection generated

$$493 \times 8 = 3,944 \text{ samples}. \quad (19)$$

B.2.4 Type B inventory

The Type B condition contains 200 prompts (`configs/prompts_typeB_v1.json`) and a frozen reference set (`configs/typeB_reference.json`). Type B items are real entities in the same five domains as Type A (40 per domain as a target, total 200 binding). Phase 2 applies the same eight-sample protocol; Type B is a secondary analysis and is not used to size the Type A mixed-prompt inventory.

B.3 Sampling and phase transition

B.3.1 Phase 1

Phase 1 begins with the 1,000-prompt Type A inventory. For every prompt, 8 generated samples are collected:

$$N_{\text{Phase1}} = 1,000 \times 8 = 8,000. \quad (20)$$

The resulting pilot set is stored as `results/phase1_pilot_trajectories.json`, and the freeze record indicates that the corresponding generation rule passed the specified Rule 3 check.

B.3.2 Estimation of the Phase 2 requirement

The freeze record reports: mixed prompt rate = 0.599; mixed positive rate = 0.471; calculated requirement $M = 444.1$; rounded requirement = 445; independent Phase 2 inventory = 493 prompts. The Phase 2 inventory was therefore fixed at 493 prompts before Phase 2 generation proceeded.

B.3.3 Phase 2 generation

The complete Phase 2 generation consists of

$$493 \text{ prompts} \times 8 \text{ samples} = 3,944 \text{ generated samples}. \quad (21)$$

Labels match Table 2. Generated trajectory metadata is stored in `results/phase2_trajectories.json`, with projected layer-wise trajectory data in `results/phase2_projected/*.npz` and the designated raw hidden-state ablation subset in `results/ablation_raw_hidden_states/*.npy`.

B.4 Wikipedia-based factual-conflict filtering

B.4.1 Purpose

The Wikipedia-based filtering stage is intended to reduce contamination or ambiguity arising when a generated response conflicts with externally verifiable factual content. The filtering stage is applied before the final Phase 3 mixed-prompt evaluation set is established. The reported Phase 3 result states that, after the specified mixed-prompt filtering procedure, 299 prompts and 2,392 samples remained, consistent with

$$299 \times 8 = 2,392. \quad (22)$$

B.4.2 Conceptual procedure

At the level supported by the present record, the filtering process is

$$\begin{aligned} &\text{generated response} \rightarrow \text{identify factual / reference-bearing content} \\ &\rightarrow \text{Wikipedia conflict check} \rightarrow \text{apply frozen filtering / exclusion rules} \\ &\rightarrow \text{retain eligible mixed prompts / samples} \rightarrow \text{Phase 3 evaluation set.} \end{aligned} \quad (23)$$

The exact implementation details are normative experimental content and must be taken from the frozen filtering source.

B.4.3 Implementation (frozen protocol)

Type A entity collision is checked with the Wikipedia MediaWiki API (`action=query`), using the missing flag to detect titles that do not resolve to an existing page [Nakamura, 2026b]. Lookup is title/entity based, not a learned similarity threshold. A second backend (e.g. Google Custom Search) is *not* used. Entities absent from Wikipedia (minor toponyms, non-English names) can therefore be misclassified as fictional; that residual risk is logged and is not described as complete verification. Normalization for Type B reference matching is lowercasing, whitespace stripping, and punctuation stripping, with numeric tolerance recorded in `configs/typeB_reference.json` together with `configs/refusal_patterns_en.json`. Conflict is a prompt-construction / eligibility control, not a token-level label. Execution order is

$$\begin{aligned} \text{frozen prompt inventory} &\rightarrow \text{generation (8 samples, temperature 0.8)} \\ &\rightarrow \text{Wikipedia collision check (Type A)} \rightarrow \text{label assignment} \\ &\rightarrow \text{hidden-state extraction} \rightarrow \text{mixed-prompt eligibility} \rightarrow \text{hashed split.} \end{aligned} \quad (24)$$

B.5 Mixed-prompt filtering

The Phase 3 record reports: Phase 2 prompt pool 493; Phase 2 generated samples 3,944; retained mixed prompts after filtering 299; retained mixed samples 2,392. The reduction is

$$493 \rightarrow 299 \text{ prompts,} \quad 3,944 \rightarrow 2,392 \text{ samples,} \quad (25)$$

with prompt-retention fraction

$$\frac{299}{493} \approx 0.6065. \quad (26)$$

Because the available record identifies the result of the filtering stage but does not expose all implementation rules, this fraction should be treated as an empirical outcome, not as a design target.

B.6 Prompt identity, determinism, and provenance

Prompt identities are part of the experiment’s reproducibility boundary. The freeze manifest records explicit hashes for `configs/prompts_v1.json`, `configs/prompts_v1_phase2.json`, `configs/calibration_prompts_v1.json`, `configs/prompts_typeB_v1.json`, `configs/typeB_reference.json`, and `configs/refusal_patterns_en.json`.

The data split itself is also hashed (`configs/data_splits.json`) with the recorded Phase 2 split: train 296, validation 99, test 98.

The designated ablation prompt inventory is separately hashed (`configs/ablation_prompt_ids.json`); the record explicitly states that 98/200 designated ablation prompts were available, with the shortfall retained in the file’s `shortfall` field.

For a future exact reproduction, the correct procedure is therefore: (i) obtain the frozen prompt assets; (ii) verify their SHA-256 hashes; (iii) reproduce the prompt IDs and split assignments; (iv) apply the frozen filtering implementation; and (v) verify the retained counts before executing any downstream hidden-state analysis.

B.7 Relationship between prompt-level and sample-level filtering

The experiment distinguishes prompt inventories from generated samples. A single prompt generates multiple independent samples:

$$p_i \rightarrow \{y_{i,1}, y_{i,2}, \dots, y_{i,8}\}. \quad (27)$$

A prompt-level decision and a sample-level decision are not interchangeable. The reported Phase 3 record retains 299 prompts and 2,392 samples, exactly 8 samples per retained prompt. To preserve balanced variance estimation across generations, eligibility is evaluated at the prompt-cluster level: a prompt is retained if and only if its eight generated samples meet the mixed-response criterion (not all-positive and not all-negative), so no partial sample clusters enter Phase 3.

Table 12: Reproducibility status of protocol components. Filenames are relative to configs/.

Protocol component	Frozen source	Hash / version	Status
Calibration prompts	calibration_prompts_v1.json	freeze manifest	Verified
Type A Phase 1 prompts	prompts_v1.json	freeze manifest	Verified
Type A Phase 2 prompts	prompts_v1_phase2.json	freeze manifest	Verified
Type B prompts	prompts_typeB_v1.json	freeze manifest	Verified
Type B references	typeB_reference.json	freeze manifest	Verified
Refusal / hedging patterns	refusal_patterns_en.json	freeze manifest	Verified
Split assignment	data_splits.json	freeze manifest	Verified
Ablation prompt IDs	ablation_prompt_ids.json	freeze manifest	Verified (short-fall)
Wikipedia collision check	MediaWiki (missing)	action=query plan v1.4	Specified
Prompt templates (Type A)	five fictional-entity templates	plan v1.4	Specified

B.8 Label assignment and filtering order

The frozen record separately identifies generation configuration, refusal / hedging patterns, hallucination label logic, Type B reference rules, and data filtering. For reproducibility, the final manuscript should explicitly state the order in which these operations occur.

The manuscript currently supports the following high-level dependency:

$$\begin{aligned}
 &\text{prompt inventory} \rightarrow \text{generation} \rightarrow \text{generated samples} \\
 &\rightarrow \{\text{factual / Wikipedia filtering, label determination, hidden-state trajectory extraction}\} \quad (28) \\
 &\rightarrow \text{Phase 3 eligibility / evaluation set.}
 \end{aligned}$$

The frozen execution order is the pipeline in §B.4: inventory, generation, Wikipedia collision check, label assignment, trajectory extraction, mixed-prompt eligibility, hashed split.

B.9 Required reproducibility table

B.10 Audit trail

The filtering and prompt-construction appendix should be auditable from the reported counts. At minimum, the final audit trail should permit the reader to verify

$$1,000 \times 8 = 8,000 \quad (29)$$

for the Phase 1 pilot,

$$493 \times 8 = 3,944 \quad (30)$$

for Phase 2, and

$$299 \times 8 = 2,392 \quad (31)$$

for the final Phase 3 mixed set. The published record should also connect each transition to a concrete artifact hash and, where applicable, to a deterministic script or implementation module.

B.11 Scientific interpretation boundary

The Wikipedia filtering stage is a *data-quality and eligibility-control mechanism*, not itself evidence that the resulting labels are universally correct. Accordingly: the filtering procedure defines which observations enter the analysis; the label procedure defines how retained observations are categorized; the hidden-state trajectory procedure defines which internal representation is measured; and the classifier defines how trajectory-derived features are converted into predictions. These are separate methodological layers and should remain separate in the paper.

This separation is particularly important because the central scientific claim of EXP-TRJ001 concerns the geometry and predictive utility of hidden-state trajectories, not the validity of Wikipedia as a universal ground-truth source.

Table 13: Layer-level test AUROC for BL-LEN, BL-NORM, PROP- κ , and PROP- κ + LEN.

Layer	BL-LEN	BL-NORM	PROP- κ	PROP- κ + LEN
8	0.8077	0.8507	0.7090	0.8083
16	0.8077	0.8588	0.7806	0.8293
24	0.8077	0.7158	0.7167	0.8314

Table 14: Paired Layer 16 comparison against BL-LEN (98-prompt test set).

Condition	AUROC	Δ vs. BL-LEN	p
PCA 64D	0.7806	−0.0271	0.8685
PC4-67	0.7969	−0.0108	0.6935
BL-LEN	0.8077	—	—
Raw 3584D features	0.8472	+0.0395	0.004
BL-NORM	0.8588	+0.0511	—

C Detailed AUROC tables

C.1 Layer-level results

C.2 Raw and PCA ablations

C.3 Confidence intervals

Table 15: Bootstrap 95% confidence intervals for the principal conditions.

Condition	AUROC	95% CI
PCA 64D, Layer 16	0.7806	[0.7298, 0.8232]
PC4-67	0.7969	[0.7498, 0.8390]
BL-LEN	0.8077	[0.7632, 0.8475]
Raw 3584D-derived features	0.8472	[0.8060, 0.8824]
BL-NORM	0.8588	[0.8239, 0.8903]